

Deepak Mallampati

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Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who ships customer-facing AI products end-to-end in regulated domains - healthcare RAG systems with ~35% retrieval precision gains, agentic LLM workflows with >30% hallucination reduction, schema-validated multi-agent claims pipelines, and FastAPI/Docker production services with ~30% fewer post-deployment defects. Solo-shipped Manga Lens on the Chrome Web Store with four AI vision providers. Comfortable owning customer engagements from concept through production in HIPAA-conscious and compliance-sensitive environments - directly transferable to legal-domain AI deployment.

SKILLS

Customer-Facing AI Delivery • Forward Deployed Engineering • RAG Pipelines & Grounding • Agentic LLM Workflows • Multi-Provider LLM Integration • FastAPI / Docker Production • Regulated-Domain Compliance • Schema-Validated Contracts

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: LLM application development, Retrieval-Augmented Generation (RAG), agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, responsible AI

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting, SaaS Healthcare)

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval and healthcare documentation search; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~35% and reduced irrelevant retrieval by >30%. Retrieval-grounded response alignment exceeded 90% in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall by 15-20% on high-risk categories while holding precision >90%.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs - directly transferable to legal compliance contexts.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India (Oil & Gas, Enterprise ERP)

- Optimized SQL and T-SQL stored procedures powering the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing); reduced query execution time ~20% and data retrieval latency ~25%.
- Built CI/CD pipelines with Jenkins for schema updates, version-controlled SQL scripts, and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed performance dashboards using SQL Server DMVs and Grafana to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~25%.
- Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive oil & gas environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India (Nonprofit Tech, Video Analytics, Applied NLP)

- Built a transformer-based video summarization and highlight-generation system across 5,000+ recorded sessions; reduced manual review time by 60-70% (hours to <5 minutes per video).
- Implemented document Q&A with RAG (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.
- Deployed the stack as a Flask-based API with a lightweight web interface for non-technical staff.

Kent State University

Aug 2023 - May 2025

Student Manager - Ohio, USA (Higher Education, Operations Management)

- Led coordination for 150+ students across academic and extracurricular initiatives; reduced coordination delays ~30% and logistical issues ~25%.
- Formalized onboarding for new team members, cutting ramp-up time ~20%.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, and captureVisibleTab for panel capture. Integrates four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline handles tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 manga/webtoon sites, privacy policy, and narrowed host permissions.

Direct solo-ship proof for founding-FDE customer-facing delivery.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization -> Validation -> Consistency Analysis -> Duplicate Detection -> Fraud Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces - same architectural pattern transfers to legal contract validation and compliance audits.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app that interprets dreams, generates poetic reinterpretations, and synthesizes dreamscapes via coordinated multimodal APIs (Perplexity Sonar, Replicate). Introduced intermediate structured prompt transformation layers - improved contextual alignment ~30% and first-pass image success rate ~25-30%.

E-Farming Digital Marketplace (Founder & Full-Stack Developer)

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded 80-120 active users in phase one; cart, reviews, and community blogging features to drive repeat engagement - founder-pattern evidence for founding-team scope.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025